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PAPER

Enhancement of quaternary nitrogen doping of graphene oxide *via* chemical reduction prior to thermal annealing and an investigation of its electrochemical properties†

Tran Ngoc Huan,‡[a] Tran Van Khai,‡[b] Youngjong Kang,[a] Kwang Bo Shim[b] and Hoeil Chung*[a]

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A simple and efficient method to enhance the quaternary nitrogen doping (N-doping) of graphene has been demonstrated. Recent studies have shown that quaternary N in the graphene network provides more efficient electrocatalytic activity. Therefore, a novel strategy to enhance the quaternary N-doping is currently in high demand. The strategy employed in this work was to modify graphene oxide (GO) prior to thermal annealing so as to provide a more efficient structure for quaternary N doping. GO was first chemically reduced with hydrazine to substantially increase the formation of C=C bonds and simultaneously decrease the atomic oxygen concentration. The reduced graphene oxide (RGO) was then annealed in the presence of NH_3. Although N-doping *via* the replacement of oxygen is preferred, the probability of carbon being substituted with N dopants in the graphitic structure of RGO could increase due to the relatively higher content of C=C when compared to the atomic oxygen concentration. In addition, due to the decreased atomic oxygen concentration, the electro-conductivity was enhanced. Cyclic voltammograms (CVs) of 5 mM $K_3Fe(CN)_6$ and 2 mM H_2O_2 were used to examine the electrochemical response of the quaternary N-maximized RGO. An improvement in electrocatalytic reduction and a higher electro-conductivity were confirmed based on an analysis of the obtained CVs.

1. Introduction

Previous research has shown that the electro-conductivity of graphene can be improved through nitrogen doping (N-doping). Thus, N-doping is frequently utilized to improve the performance of graphene-based electrochemical applications.[1–5] Lin's group demonstrated the N-doping of graphene by a plasma method. However, the percentage of doped nitrogen in the graphene network was only about 1.35% and a large amount of oxygen (~28%) still remained in the structure. In addition, the researchers identified the presence of three different types of nitrogen: pyridinic N, pyrrolic N, and quaternary N.[1] Alternatively, Dai's group reported on a thermal annealing method that enabled N-doping and the removal of oxygen simultaneously. The method increased the content of doped nitrogen up to 4.0% and substantially decreased the oxygen concentration to 4.2%. The group also showed that the content of atomic oxygen in the graphene was an important factor governing its electro-conductivity.[6,7]

Recent studies have shown that the efficiency of the oxygen reduction reaction (ORR) at the surface of graphene could vary depending on the relative concentration of three types of nitrogen out of the total doped nitrogen concentration.[8–11] Based on an analysis using ultraviolet photoemission spectroscopy (UPS) and the electrochemical method, Lin *et al.* found that the ORR activity was not proportional to the total amount of doped nitrogen and pyridinic N was not effective in promoting the activity of ORR.[8] Furthermore, Oshima *et al.* demonstrated that sp^2 hybrid graphite-like nitrogen, such as quaternary N, was more efficient with respect to the electrocatalytic activity.[9] It was also found that pyridinic N did not induce a significant increase in the ORR activity. An improvement in the ORR activity of carbon alloy catalysts was demonstrated when these materials were prepared with a higher concentration of quaternary N. Consequently, an N-doping strategy that can selectively increase the percentage of quaternary N has become more valuable when graphene is utilized for fuel cells and electrochemical sensors.[9–11] The attainment of good electro-conductivity is also a crucial issue in the application of graphene. In this work, a simple and effective N-doping method to enhance the quaternary N doping and simultaneously decrease the atomic oxygen concentration in graphene is proposed. The strategy was to modify graphene

[a]*Department of Chemistry, College of Natural Sciences, Hanyang University, Seoul, Korea. E-mail: hoeil@hanyang.ac.kr*
[b]*Department of Materials Science and Engineering, Hanyang University, Seoul, Korea*

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‡ These authors contributed equally to this work.

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oxide (GO) prior to thermal annealing so as to induce a more efficient structure and enhance the quaternary N doping. GO was first synthesized according to Hummers' method and then chemically reduced with hydrazine.[12–14] The goal of the chemical reduction process was to increase the formation of C=C bonds in the GO, which could be efficient doping sites for quaternary N in the subsequent thermal annealing step; this has not been demonstrated in previous studies. It was also possible to decrease the atomic oxygen concentration so as to improve the electro-conductivity. After the chemical reduction, the reduced graphene oxide (RGO) was annealed at high temperatures in the presence of NH_3 for N-doping. For a comparison, GO was directly annealed under the same conditions as those employed in previous studies.[6] X-ray photoelectron spectroscopy (XPS) was used to identify three different types of nitrogen and determine the atomic composition of the samples. In addition, the sheet resistance of the samples was measured to examine the variation in the electro-conductivity according to the annealing conditions. The electrochemical characteristics of RGO, N-doped GO, and N-doped RGO were examined. For this analysis, each sample was individually prepared on the surface of a Pt electrode. Cyclic voltammograms (CVs) of a $K_3[Fe(CN)_6]$ solution were then acquired using each electrode. These electrodes were further used to measure the different concentrations of H_2O_2. The resulting electrochemical responses, such as the peak shape and reduction potential, were compared.

2. Experimental

Graphene oxide (GO) was initially synthesized by Hummers' method. The GO was then chemically reduced by immersing the material in a pure hydrazine solution for one day at 95 °C with gentle stirring. The RGO was subsequently cleaned using distilled water under sonication until the pH reached 7.0. Pt was used as a substrate for the thermal annealing due to its high melting temperature. Fourteen different high purity (99.99%) Pt substrates with a diameter of 1 mm were used. Before the thermal annealing, a solution containing RGO was spray-coated on the surface of the Pt. The RGO was annealed in a horizontal quartz tube furnace by varying the temperature from 600 to 900 °C at intervals of 100 °C in the presence of NH_3. For control of the NH_3 concentration, Ar was used as a make-up gas.

Before the thermal annealing treatment, the temperature of the furnace was set to a target temperature and high purity Ar gas was introduced into the furnace at a flow rate of 80 sccm (standard cubic centimeters per minute) so as to eliminate oxygen (O_2). A desired NH_3 concentration was then introduced into the chamber for N-doping. Annealing was carried out for 15 minutes for each case. For a comparison, GO was directly annealed without prior chemical reduction under the same experimental conditions. Images of the GO, RGO, and annealed samples were obtained with a high-resolution transmission electron microscope (HR-TEM, Tecnai 20, Hillsboro, OR, USA) operated at 200 kV. Atomic force microscopy (AFM) images were obtained with an XE-100 AFM (Park Systems Corp., Suwon, Korea). XPS spectra (ESCALAB 220, VG Microtech, UK) of the samples were collected with a Mg Kα X-ray source. Raman spectra were obtained with a dispersive Raman microscope equipped with a 532 nm laser (power: 2.9 mW, JASCO, Germany). All electrochemical measurements in this study were performed using a standard cell with a three-electrode arrangement; Ag/AgCl and Pt wires were used as the reference and counter electrodes, respectively. Data were recorded using a potentiostat (eDAQ, Model PowerLab/4SP, Australia).

3. Result and discussion

Characterization of GO and RGO

A TEM image of GO prepared by Hummers' method is shown in Fig. 1(a). As evident in the figure, a clear sheet-form of GO was synthesized; several locally folded sections are also observed. Then, the corresponding AFM image (4 × 4 μm) covering several GO sheets was acquired as shown in Fig. 1(b) and the thicknesses of two GO sheets were measured. The inset denotes the variation in the thicknesses across the corresponding line mappings (designated as a line on the AFM image). Based on the observation, the thickness of the graphene was approximately 1.1–1.2 nm.

Both as-synthesized GO and RGO were initially characterized *via* X-ray photoelectron spectroscopy (XPS). The corresponding

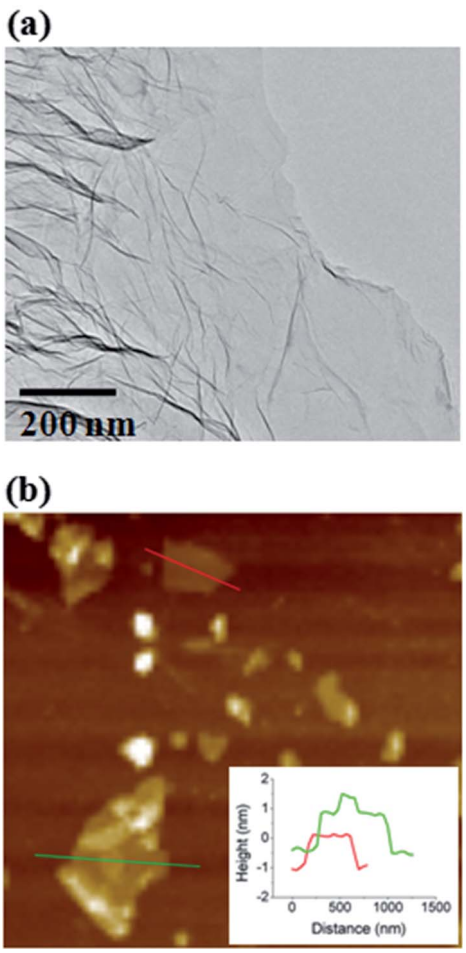

Fig. 1 TEM image of GO (a) and AFM image covering several GO sheets (b). The inset in (b) corresponds to the variation in the thicknesses across the corresponding line mappings (designated as a line on individual GO images) of two GO sheets.

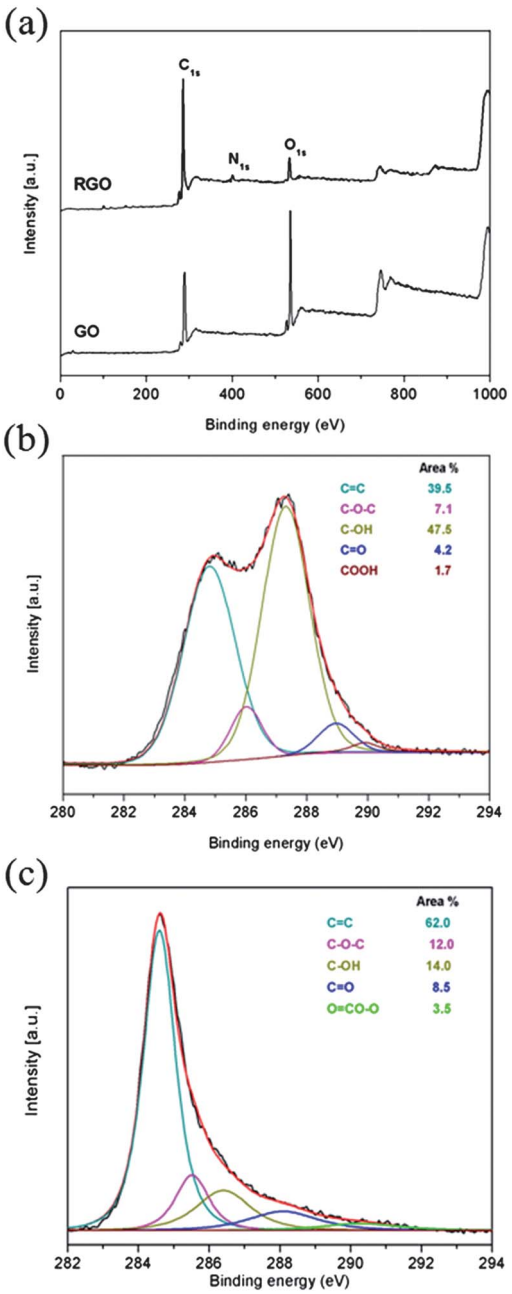

Fig. 2 (a) Survey XPS spectra of GO and RGO in the 0–1000 eV BE range. (b) High-resolution C spectra of GO and (c) high-resolution C spectra of RGO. The peaks under both C peaks are found by a curve-fitting method. Each Gaussian peak is denoted with a different color and the corresponding types of carbon bonding are designated in each figure.

survey spectra in the 0–1000 eV binding energy (BE) range are shown in Fig. 2(a). After the chemical reduction of the GO, the intensities of the C and O peaks substantially increased and decreased, respectively. The atomic compositions of the GO and RGO calculated from the corresponding XPS spectra are summarized in Table 1. The elemental concentrations were determined by calculating the relative percentage of each peak area to the total peak area of the bands.[6] After the reduction process, the oxygen concentration significantly decreased from 31.20 to 14.65%, while the concentration of carbon increased from 68.80 to 84.52%.

Table 1 Calculated atomic compositions of GO and RGO

Sample	C (%)	N (%)	O (%)
GO	68.80	0.00	31.20
RGO	84.52	0.83	14.65

To investigate GO structural variations induced by the chemical reduction process, high-resolution XPS spectra of C in the BE range of 280 to 294 eV were obtained. Using the acquired peaks, a curve-fitting method was applied so as to resolve the peaks into individual Gaussian components and identify different types of carbon bonding. The C spectrum of GO and 5 different peaks found by the curve-fitting are shown in Fig. 2(b). Each Gaussian peak is denoted with a different color. The type of carbon bonding and relative percentage of the peak area are also summarized using the sample color notation. The peaks at 284.7, 286.0, 287.0, 288.6, and 289.8 eV correspond to the sp^2 carbon bond (C═C, dark cyan), sp^3 carbon in C–O–C (magenta), C–OH (dark yellow), C═O (blue), and O═C–OH (wine), respectively. The C spectrum of RGO and 5 different peaks found by the curve-fitting are shown in Fig. 2(c). The type of carbon bonding and relative percentage of the peak area are also displayed. The shape of the C peak from RGO is significantly different from that observed for GO. Such a finding clearly indicates that a considerable variation in the carbon bonding structure occurs in GO as a result of the chemical reduction process. The peaks in Fig. 2(c) are assigned to C═C (dark cyan), C–O–C (magenta), C–OH (dark yellow), C═O (blue), and O═CO–O (green), respectively. The percentage of C═C increased from 39.5 to 62.0%, as confirmed by the narrow and strong C═C peak in the C spectrum of RGO. In contrast, the percentage of C–OH significantly decreased from 47.5 to 14.0% after the reduction process. Relatively minor variations in the percentage of C–O–C and C═O are also observed. The overall C/O atom ratio for GO before and after the chemical reduction process changed from 2.26 to 5.77, respectively. Consequently, the chemical reduction of GO induces a substantial increase in the presence of C═C and a simultaneous decrease in the atomic oxygen content. It was our expectation that the variation in the carbon bonding structure in RGO, especially the increase in the percentage of C═C, could provide efficient sites for quaternary N doping when RGO is annealed in the presence of NH_3.

Determination of NH_3 concentration for optimal N-doping of RGO

N-doping of RGO was performed by thermally annealing the material in the presence of NH_3. Therefore, the concentration of NH_3 that yields optimal N-doping must first be determined. For this purpose, the annealing temperature was initially set at 900 °C based on a report from Dai's group.[6] Such a temperature was found to maximize N-doping and yield the highest conductivity. With the annealing temperature maintained at 900 °C, N-doping was performed by changing the concentration of NH_3 from 5 to 20% in increments of 5%. The sheet resistance of the N-doped samples was measured to examine the effect of N-doping under each condition, since this property is directly related to the degree of N-doping and the concentration of oxygen in the structure. For

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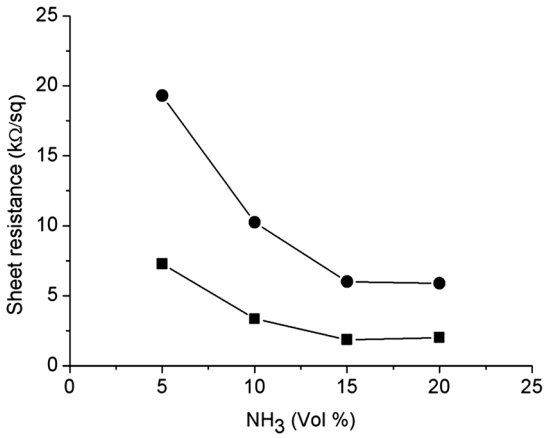

Fig. 3 Variation in the sheet resistance for both N-doped GO (circles) and N-doped RGO (squares) as a function of the NH_3 concentration. Both samples were annealed at 900 °C.

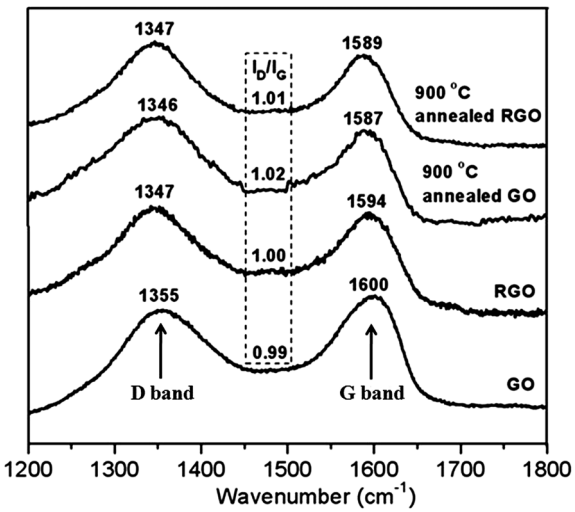

Fig. 4 Raman spectra (λ_{ex} = 532 nm) collected from GO, RGO, N-doped GO, and N-doped RGO. N-doping was performed at 900 °C. I_D/I_G (intensity ratio of D-bands to G-bands) values are also shown for each spectrum.

a comparison, GO was also annealed under identical conditions. Temperatures over 900 °C were not investigated in this study due to the possibility of thermal breakage of the C–N bond.[6]

The variation in the sheet resistance for both N-doped GO and N-doped RGO as a function of the NH_3 concentration is shown in Fig. 3. In both cases, the sheet resistance decreases with an increase in the NH_3 concentration; the variation in the sheet resistance becomes insignificant above 15% NH_3. The decrease in the sheet resistance is a consequence of an increase in the doped nitrogen content and the removal of atomic oxygen. At each NH_3 concentration, the sheet resistance of the N-doped RGO is always lower than that of the N-doped GO because the oxygen concentration in the RGO is substantially lower before the thermal annealing (see Table 1). The results provide preliminary confirmation that the chemical reduction of GO prior to N-doping, rather than direct N-doping into GO, could increase the electro-conductivity of the material. Consequently, the concentration of NH_3 was maintained at 15% in subsequent experiments.

To further examine the degree of N-doping, Raman spectra (λ_{ex} = 532 nm) of both N-doped GO and N-doped RGO at 900 °C were obtained; the results are shown in Fig. 4. For a comparison, the Raman spectra of GO and RGO are also displayed in the figure. G-bands (corresponding to the optical mode vibration of two neighboring carbon atoms) and D-bands (corresponding to defects in graphene) were observed for all samples.[15–17] The G-bands of both GO and RGO shift to a lower wavenumber after the N-doping, as is consistent with the findings of previous research. For both materials, the intensity of the D-band increases relative to that of the G-band after the N-doping due to an increase in the defects. To verify the obtained findings, I_D/I_G (ratio of the intensity of the D-bands (I_D) to the intensity of the G bands (I_G)) values are also shown in the figure. The observed Raman spectral features could additionally indicate the variation of GO structure with N-doping.[18–20,24]

Variation of the degree of quaternary N-doping at different annealing temperatures

As stated above, an increase in the concentration of quaternary N in graphene could facilitate better catalytic performance. In addition, the minimization of the atomic oxygen concentration leads to improved electro-conductivity. Therefore, the variation in the relative percentage of quaternary N-doping over the total N-doping and the electro-conductivity as a function of the annealing temperature were investigated. For this purpose, both GO and RGO were annealed from 600 to 900 °C in increments of 100 °C, while the concentration of NH_3 was kept constant at 15%. After the annealing at different temperatures, XPS spectra for all N-doped samples were obtained. Survey XPS spectra of N-doped GO and N-doped RGO at different temperatures in the BE range of 0 to 1000 eV are shown in Fig. 5(a) and (b), respectively.

In both cases, an N peak clearly appears in each spectrum after annealing in the presence of NH_3. This peak is absent in the spectrum of GO and substantially weak in the spectrum of RGO (see Fig. 2(a)). Such a result clearly indicates the incorporation of nitrogen (N-doping) into the samples. Meanwhile, no systematic variation in the intensity of the N peak with respect to the annealing temperature is observed for both cases. In contrast, the intensity of the O peak in the spectra of N-doped GO decreases as the annealing temperature is increased, indicating a lower atomic oxygen concentration at higher temperatures. The decrease in the O peak with a temperature increase is also observed in the spectra of N-doped RGO, although the intensity variation is marginal.

The atomic concentrations of C, N, and O calculated from the XPS spectra in Fig. 5 for both N-doped GO and N-doped RGO at different temperatures are listed in Table 2. In both cases, the C and O concentrations increase and decrease, respectively, as the annealing temperature is increased. When RGO is annealed at 900 °C, the concentrations of C and O reach a maximum and minimum, respectively. The minimization of the O concentration is clearly responsible for the lowest sheet resistance obtained under this condition (see Fig. 3). Meanwhile, no systematic variation in the N concentration according to the annealing temperature was observed.

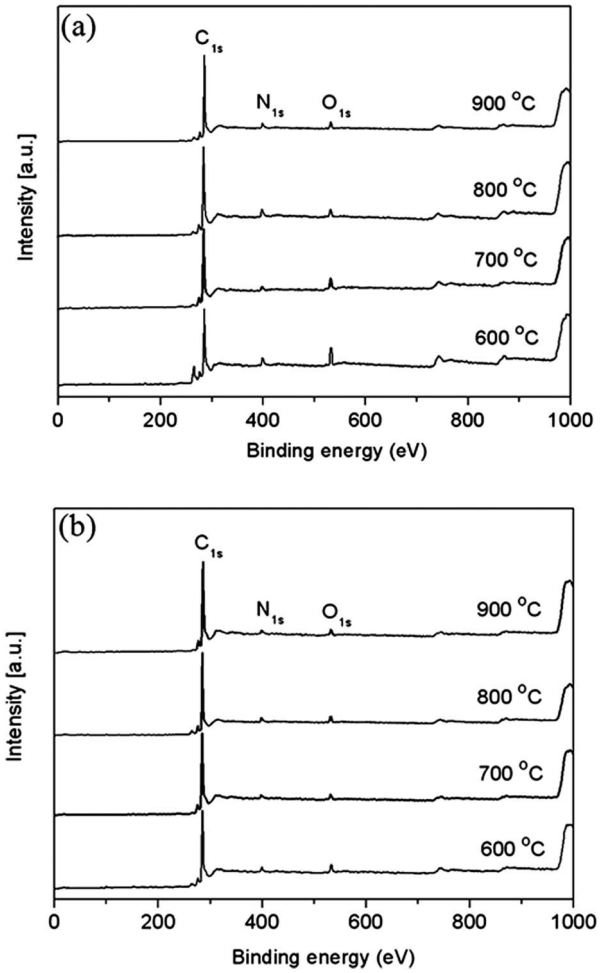

Fig. 5 Survey XPS spectra (BE range of 0 to 1000 (eV)) for (a) N-doped GO and (b) N-doped RGO at different annealing temperatures.

Table 2 Calculated atomic concentrations of C, N, and O for both N-doped GO and N-doped RGO at different temperatures

Sample	Annealing temp. (°C)	C (%)	N (%)	O (%)
GO	600	87.45	5.45	7.10
	700	90.45	3.90	5.65
	800	91.28	4.26	4.46
	900	91.61	4.34	4.05
RGO	600	90.20	4.72	5.08
	700	91.07	4.97	3.96
	800	92.77	4.12	3.11
	900	93.30	4.02	2.68

To quantitatively analyze the concentrations of three types of nitrogen (pyridinic N, pyrrolic N, and quaternary N), high-resolution XPS spectra in the N binding energy (BE) range (390–410 eV) were collected and a curve-fitting method was used to identify the three species. High-resolution XPS spectra of N obtained from GO annealed at 700, 800, and 900 °C are shown in Fig. 6(a–c), respectively. At 700 °C, only two peaks corresponding to pyridinic N (blue) and pyrrolic N (magenta) are observed. The peak of quaternary N (dark yellow) starts to appear at an annealing temperature of 800 °C and its magnitude

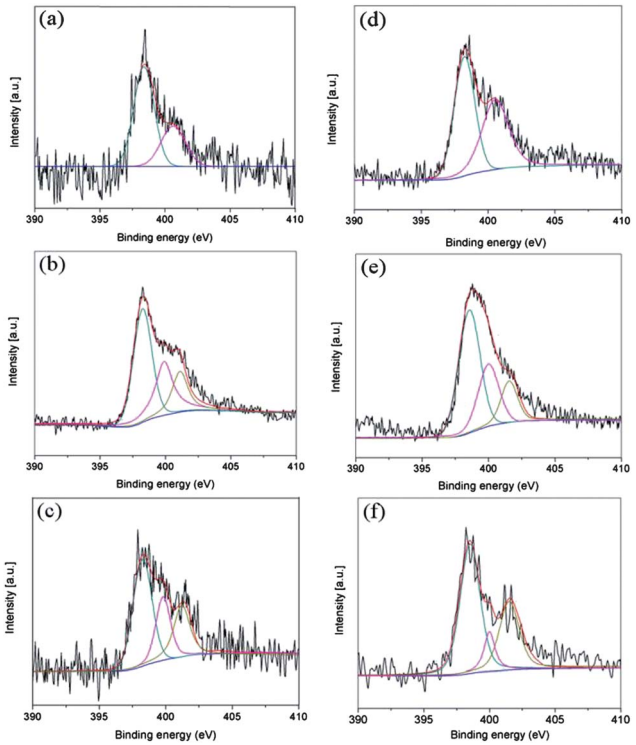

Fig. 6 High-resolution N_{1s} spectra for N-doped GO at annealing temperatures of (a) 700, (b) 800, and (c) 900 °C, and N-doped RGO at annealing temperatures of (d) 700, (e) 800, and (f) 900 °C. The Gaussian peaks correspond to three different types of nitrogen (pyridinic N: blue, pyrrolic N: magenta, quaternary N: dark yellow) found by a curve-fitting method.

becomes larger at 900 °C. XPS spectra of N acquired from RGO annealed at 700, 800, and 900 °C are shown in Fig. 6(d–f), respectively. Similar to the findings obtained for GO, the peak of quaternary N appears at 800 °C and its intensity increases at 900 °C. Based on the observations from the collected XPS spectra, the concentration of quaternary N increases when either GO or RGO is annealed at elevated temperatures.

Fig. 7 shows the variation in the quaternary N concentration relative to the total N-doped concentration as a function of the annealing temperature for both GO (circles) and RGO (squares). The error bars correspond to the standard deviations of concentrations obtained from 3 separately annealed samples. The concentrations of quaternary N are quite similar up to 800 °C for both cases. However, the quaternary N content substantially increased when RGO was annealed at 900 °C. The increase in quaternary N is accompanied by a decrease in pyrrolic N, as shown in Fig. 6(c) and (f). Based on the findings of previous research, N-doping into the graphene structure preferably proceeds through the replacement of oxygen in GO.[6] Since the atomic oxygen concentration is substantially lower and the content of C=C is higher in RGO when compared to GO, the probability of carbon being substituted with nitrogen in the graphitic structure would increase, although the replacement of oxygen by nitrogen would be still preferable. The increased chance for the substitution of carbon in the graphitic structure by nitrogen could explain the relatively higher concentration of quaternary N after annealing RGO at 900 °C. It should be noted

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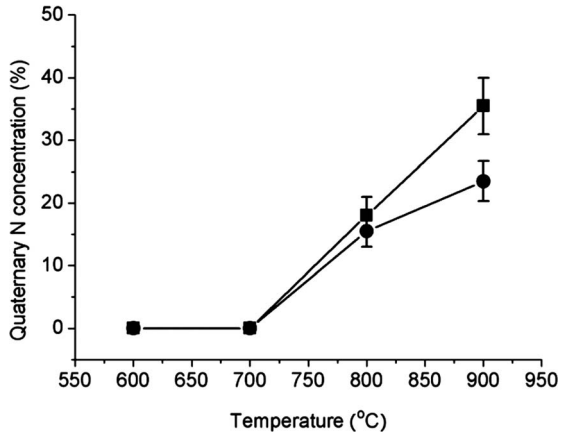

Fig. 7 Variation of the quaternary N concentration for both GO (circles) and RGO (squares).

that the sums of the carbon and oxygen concentrations at four different temperatures remain quite constant, 95.51 ± 0.67% and 95.40 ± 0.33% for the annealing of RGO and GO, respectively. Such results could support the scenario of N-doping by simultaneously replacing oxygen and carbon in the graphitic structure. Consequently, the chemical reduction of GO prior to thermal annealing is effective for enhancing the quaternary N-doping. Alternatively, IR spectroscopy was used to provide supplementary information to investigate the quaternary N-doping as shown in the ESI†.

Examination of electrochemical responses from N-doped RGO

To examine the electrochemical response of N-doped RGO, a Pt electrode covered with RGO annealed at 900 °C (referred to as a 900 °C RGO-Pt electrode) was prepared and used to collect cyclic voltammograms (CVs) of 5 mM $K_3[Fe(CN)_6]$ in the range of −200 to 800 mV at a scan rate of 5 mV s^{-1}. CVs were also acquired for the same sample using a glassy carbon electrode (GCE), a RGO-Pt electrode, and a 900 °C GO-Pt electrode for the purpose of comparison. Electrochemical measurements were not obtained for GO since its conductivity was too low. All 4 electrodes had a diameter of 1 mm. The resulting CVs are shown in Fig. 8(a). In all cases, the CVs exhibit a typical duck-shape with no indication of an unusual electrochemical response. The intensity of the redox peaks is strongest when the 900 °C RGO-Pt electrode is employed. The maximum peak intensity could be attributed to the lowest sheet resistance of the 900 °C-annealed RGO (Fig. 3), which is in turn a consequence of the lowest oxygen concentration in the structure.

To investigate the electrocatalytic reduction performance, CVs were collected for 2 mM H_2O_2 in a PBS buffer as in previous research[1,5,21–23] using the same 4 electrodes used to obtain the data in Fig. 8(b). For the purpose of comparison, a bare Pt electrode was used to measure the same solution. In each case, the initial reduction potential of H_2O_2 is different. Reduction potentials of −0.25, −0.10, 0.08, 0.11, and 0.15 V were obtained for GCE, RGO-Pt, 900 °C GO-Pt, 900 °C RGO-Pt, and bare Pt electrodes, respectively. The initial reduction potential acquired from the Pt electrode is lower when compared to that from the 900 °C RGO-Pt electrode. However, the current density obtained

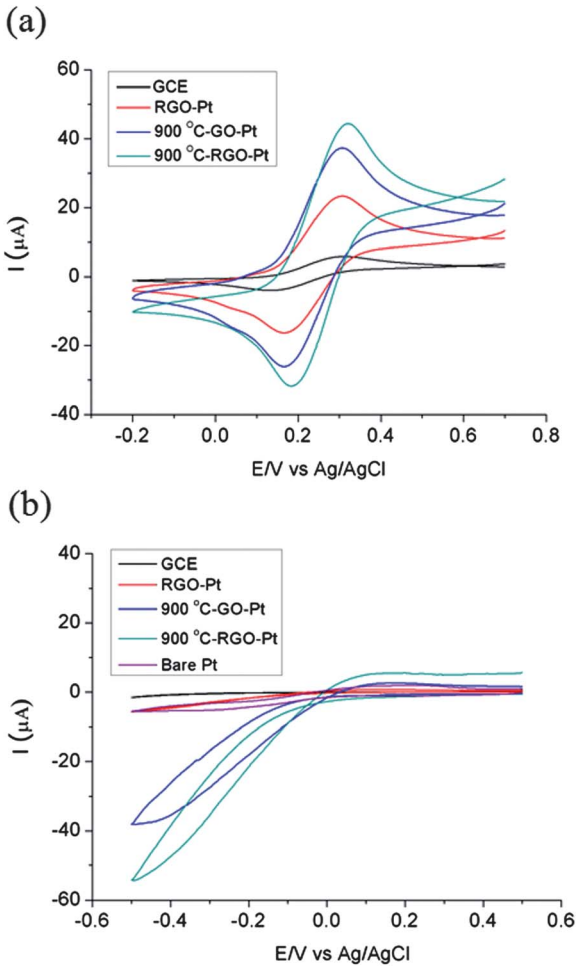

Fig. 8 (a) Cyclic voltammograms (CVs) of 5 mM $K_3[Fe(CN)]_6$ acquired with 4 different electrodes (GCE, RGO-Pt, 900 °C GO-Pt, and 900 °C RGO-Pt). (b) CVs of 2 mM H_2O_2 acquired with 5 different electrodes (GCE, RGO-Pt, 900 °C-GO-Pt, 900 °C-RGO-Pt, and bare Pt).

with the 900 °C RGO-Pt electrode is substantially higher in the potential range of 200 to −500 mV. In addition, when compared to the results obtained with the 900 °C GO-Pt electrode, the initial reduction potential is lower when the 900 °C RGO-Pt electrode is used for the measurements. These findings indicate that the electrocatalytic reduction of H_2O_2 is improved with 900 °C annealed RGO possessing a larger quaternary N concentration in the graphene structure.

4. Conclusion

The obtained results demonstrate that the chemical reduction of GO before N-doping by thermal annealing efficiently increased the relative concentration of quaternary N in the graphene structure and enhanced the electro-conductivity of the material. In addition, the synthesis of the structure is simple and straightforward. We believe that the proposed graphene structure has strong potential for application in fuel cells and electrochemical sensing. When target-specific biomolecules, such as antibodies or aptamers, are conjugated onto the surface of the structure, diverse bio-analytical measurements with good sensitivity could be realized.

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